The Doctor Will See You Now: Stakes, Verifiability, and the Supervision of Clinical AI

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Abstract

The debate over artificial intelligence in medicine is usually framed as a choice between a human clinician and an AI one. We argue the real choice is different. It is where a decision sits on a supervision ladder. The bottom rung is a human working alone. The middle is an AI that drafts and a human who approves. The top is an AI that acts alone. Which rung a decision belongs on is set by two properties of the decision, not by the technology. The first is how much harm a wrong answer does that is not later caught and corrected, the residual stakes. The second is how hard the answer is to verify. We build a model in which both a patient's willingness to accept AI care and a clinic's safe division of labor turn on these two primitives. Patients accept AI below a complexity threshold and want a human above it. The reason is that aversion rises where the decision is least verifiable. The clinic's best policy is to review every decision above a cutoff in the same index, which is why risk tiers look as they do. The two thresholds move together. The gap between them is where patients fear safe AI or accept unsafe AI, and it is where oversight and disclosure matter. The sharpest results come from separating accuracy from verifiability. The task to automate first is the verifiable one, not the low-stakes one. A model's accuracy can be proved only where its work can be checked. So better AI can raise the right amount of reliance on a human decision-maker for unverifiable, high-stakes decisions rather than lower it. A human in the loop collapses the gap by raising acceptance and lowering risk at once. That is the equilibrium the market, the regulator, and physician adoption have already found, while the autonomous AI doctor has failed. We close with the dynamics of earning autonomy.

1 Introduction

Ask whether people want a human doctor or an AI one, and the surveys answer plainly. A national survey finds that sixty per cent of American adults would be uncomfortable if their own physician relied on AI to diagnose them and recommend a treatment [Pew Research Center, 2023]. A peer-reviewed study finds that sixty-three per cent would not accept an AI-written medical answer unless a clinician reviewed it first. The same study finds that sixty-four per cent would wait three months for a human therapist rather than start at once with an AI one [Kim et al., 2024]. Yet over the same period the use of AI in care grew quickly. Physician adoption nearly doubled in a year [American Medical Association, 2024], and the capital and the regulatory clearances flowed steadily toward it. The discomfort and the adoption look contradictory only if the question is human or machine. They fit together once the question is the right one.

The right question is not who delivers care. It is how much of the decision a machine may make before a human must sign it. That is a ladder, not a switch. On the lowest rung a clinician

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works alone. On the next rung the AI prepares the work, such as a pre-visit brief, a draft plan, or a suggested order, and a clinician reviews and approves it before it reaches the patient. Higher up the AI handles the routine cases on its own and a human audits a sample. At the top the AI acts with no human at all. The claim of this paper is that the rung a decision belongs on is set by the decision, not by how good the AI is. The same two properties of the decision govern both what patients will accept and what a clinic can safely do.

The two properties are these. The first is the residual stakes. This is the harm a wrong decision causes that is not later caught and undone. A missed dose adjustment that the next visit corrects has low residual stakes even if the error rate is not low, because the error is reversible. A missed cancer has high residual stakes because it is not. The second property is verifiability. This is how readily anyone, patient or clinician, can tell whether the decision was right. Medical care is the textbook case of a good whose quality the buyer cannot judge even after consuming it [Arrow, 1963, Dulleck and Kerschbamer, 2006]. Verifiability falls as a decision moves from the routine and rule-bound toward the judgment-laden. We first bundle these two into a single complexity index, which carries most of the argument. We then pull them apart, because separating how good the AI is from how checkable the decision is turns out to be where the main results live.

We make five contributions. The first is a demand model. A patient chooses the modality that gives the higher utility. The patient trades the quality and price of care against an aversion to being treated by a machine. The aversion rises with complexity while the AI's quality advantage falls with it, so the patient accepts AI below a threshold and wants a human above it (Proposition 1). The aversion is not a taste the model assumes away. It is the documented resistance to medical AI, which rises precisely where patients feel their case is unique, that is, where it is least verifiable [Longoni et al., 2019, Dietvorst et al., 2015]. The second contribution adds the supervised rung. It shows that this rung widens the range of care patients accept. A human who stays accountable removes much of the aversion while catching errors, so the modality that drafts under review beats both working alone and acting alone over a wide middle (Proposition 2). The third is the supply side. A clinic that prices clinician time well reviews every decision above a cutoff and releases those below it. This is why risk tiers are a cutoff rule in the same index that governs demand (Proposition 3). The fourth ties the two together. In the maintained regime both thresholds respond to residual stakes in the same direction, so the frontier of acceptance and the frontier of safety largely coincide. The paper characterizes the wedge between them, where the two disagree, as the place for disclosure and oversight. The human in the loop collapses the wedge by moving both frontiers at once (Proposition 4). The fifth is the one we think is new, and it comes from refusing to keep stakes and verifiability bundled. A reviewer can only catch an error on a decision that can be checked. So the value of oversight splits into a detection channel that works where the task is verifiable and an accountability channel that works where it is not. This reverses the rule the severity-ordered autonomy ladders assume. The task to automate first is the verifiable one, not the low-stakes one (Proposition 5). And because accuracy is measured on verifiable tasks and then extrapolated to the rest, better AI can widen the region where patients accept automation that is not yet safe. So on unverifiable, high-stakes decisions a better model calls for keeping a human deciding rather than for less oversight (Proposition 6). We close with the dynamics of earning autonomy (Proposition 7).

The argument is deliberately AI-positive and deliberately wary. It is positive because the case for drafting under supervision is strong. A primary-care physician would need almost twenty-seven hours in a day to deliver all recommended care to a standard panel, and roughly two-thirds of that work could be carried by someone, or something, other than the physician [Porter et al., 2023]. It is wary because the model takes the failure modes seriously rather than assuming them away. Human review is not free and not perfect. A physician given a strong model did no better at diagnosis

than one without it, while the model alone did markedly better, so the human can be the binding constraint rather than the safeguard [Goh et al., 2024]. Inaccurate AI can also degrade an expert who would have been right alone [Yu et al., 2024]. A theory that wants to say where AI belongs has to hold both of these at once.

2 The supervision ladder

A clinical *task* is a single decision to be made for a patient. Refill this prescription or not, adjust this dose, order this test, make this diagnosis. We summarize a task by a scalar *complexity* $x \in [0, 1]$. It increases as the decision becomes more judgment-laden and less verifiable. Two functions of x carry the economics.

Assumption 1 (Residual stakes)**.** *A wrong decision on a task of complexity x causes harm $s(x) \geq 0$ with probability of later detection-and-correction $\phi(x) \in [0, 1]$. The* residual stakes *are*

$$\sigma(x) \;\equiv\; s(x)\,\big(1 - \phi(x)\big),$$

the expected harm that is not caught and undone. We take s nondecreasing and ϕ nonincreasing in x, so σ is strictly increasing: complex decisions are both costlier to get wrong and less likely to be reversed in time.

The residual-stakes idea is the formal version of a claim practitioners make in plain words. A regulatory sandbox application to deploy supervised AI for obesity-treatment planning, filed by the author's affiliated telehealth program and cited here with that disclosure, argues that the realistic worst case of a weak plan "is a missed improvement that the clinician catches and corrects at the next visit, not an acute emergency." It concludes that planning is therefore "a comparatively low-risk place to apply supervised AI." That is exactly σ small because ϕ is large. The error is reversible, so its residual stakes are low even though its raw stakes are not negligible.

Assumption 2 (Error rates and comparative advantage)**.** *The AI errs on a task of complexity x with probability $e(x)$ and the human with probability $e_H(x)$, both nondecreasing. The AI has the comparative advantage on routine, prediction-heavy tasks and the human on judgment-heavy ones: $e(x) - e_H(x)$ is strictly increasing, negative for small x and positive for large x.*

Assumption 2 is the prediction-versus-judgment division of labor [Agrawal et al., 2018, 2019], written as a crossing condition. It matches the evidence that models match or beat clinicians on structured routine tasks while remaining weaker where judgment dominates [Goh et al., 2024, Yu et al., 2024].

The ladder has three rungs. Under the *human* rung (H) a clinician decides alone, with error $e_H(x)$. Under the *supervised* rung (S) the AI drafts and a human reviews a fraction $\rho \in [0, 1]$ of cases. The human catches a share $d \in (0, 1]$ of the AI's errors, so the effective error is $e(x)\,(1 - \rho d)$. Under the *autonomous* rung (A) the AI decides alone with error $e(x)$, the case $\rho = 0$. The rest of the paper asks which rung a task of complexity x belongs on, from two directions.

3 Demand: the care patients accept

A patient with a task of complexity x chooses a rung to maximize utility. Receiving care at all is worth B. Each rung imposes three costs. The first is an expected harm cost equal to the harm

times the error rate. The second is a price. The third, for any rung that uses the machine, is an aversion cost. Write the harm cost as $\kappa\,\sigma(x)\,e_m(x)$ for rung m, with $\kappa > 0$:

$$U_m(x) \;=\; B \;-\; \kappa\,\sigma(x)\,e_m(x) \;-\; p_m \;-\; \pi_m(x) \;+\; \varepsilon_m, \tag{1}$$

Here p_m is the price and $\pi_m(x)$ the aversion cost. The term ε_m is an idiosyncratic taste drawn i.i.d. across rungs, which delivers choice probabilities in the logit form of McFadden [1974]. The human rung carries no aversion, so $\pi_H \equiv 0$, and p_H is the highest price. The autonomous rung is cheapest, $p_A < p_H$, and carries the full aversion.

Assumption 3 (Aversion rises with complexity)**.** *The aversion cost of autonomous AI care, $\pi_A(x)$, is nonnegative, strictly increasing in x, with $\pi_A(0)$ small. Resistance to medical AI grows as the patient perceives the case as more unique, and uniqueness is a form of unverifiability.*

Assumption 3 is an empirical regularity. Patients resist AI more as a case feels particular to them. That resistance shows up as a lower reservation price for machine care. And it is muted when the AI is seen to assist rather than replace a clinician [Longoni et al., 2019, Dietvorst et al., 2015]. We take it as a primitive and ask what it implies.

Compare the autonomous rung to the human rung. The patient prefers AI when the deterministic gap

$$\Delta V(x) \;\equiv\; \underbrace{\kappa\,\sigma(x)\,\big[e_H(x) - e(x)\big]}_{\text{AI quality advantage}} \;+\; \underbrace{(p_H - p_A)}_{\text{price saving}} \;-\; \underbrace{\pi_A(x)}_{\text{aversion}} \tag{2}$$

is nonnegative. The first term is the AI's quality edge. It is positive on routine tasks, where the AI errs less, and negative on complex ones. The second term is a positive constant. The third is increasing in x. The quality edge need not be monotone on its own, because $\sigma(x)[e_H - e]$ multiplies a rising stakes factor by a gap that starts positive and turns negative. We therefore assume the economically relevant single-crossing case: $\Delta V(0) > 0$, $\Delta V(1) < 0$, and ΔV crosses zero once. This holds whenever the rising aversion and the sign change in the quality gap together outweigh any non-monotonicity in the stakes-weighted quality term. The patient's choice is then a threshold.

Proposition 1 (The acceptance threshold)**.** *Under Assumptions 1–3 and the single-crossing condition above, the deterministic value gap $\Delta V(x)$ in (2) crosses zero once, from positive to negative, so there is a unique threshold $\hat{x}_D \in (0,1)$ with $\Delta V(\hat{x}_D) = 0$ such that patients accept autonomous AI care for $x < \hat{x}_D$ and prefer the human for $x > \hat{x}_D$; the logit probability of choosing AI, $\Lambda(\Delta V(x)/\eta)$ with scale $\eta > 0$, passes through one-half at $\hat{x}_D$. The threshold rises with the price of human care and with the AI's routine-task accuracy and falls with the level of aversion. Its comparative static in the residual stakes is signed, not automatic: since $\partial\Delta V/\partial\sigma = \kappa(e_H - e)$, higher stakes lower the threshold only in the regime $\hat{x}_D > x^\star$, where the switch to the human happens past the quality crossover $x^\star$ at which $e = e_H$; below that crossover the sign reverses and higher stakes raise the threshold. The demand story, patients refusing AI where it is more accurate, is the first regime, and we maintain it.*

Proof. By the single-crossing condition ΔV is continuous, positive at $x = 0$, negative at $x = 1$, and changes sign once, so it has a unique zero $\hat{x}_D$; the logit map Λ is increasing, so $\Lambda(\Delta V(x)/\eta)$ equals one-half exactly where $\Delta V = 0$. For the comparative statics, ΔV rises in p_H and in the AI's accuracy (which raises $e_H - e$ at low x) and falls in π_A, each shifting the crossing accordingly. The partial in stakes is $\partial\Delta V/\partial\sigma = \kappa(e_H - e)$, negative where $e > e_H$, that is above the quality crossover $x^\star$, and positive below it; so $\hat{x}_D$ falls in σ when $\hat{x}_D > x^\star$ and rises otherwise. □

Proposition 1 says the demand-side line is drawn by residual stakes and verifiability, not by the technology's average quality. A more accurate model widens the range of care patients accept. It does not erase the line, because aversion keeps rising with complexity even as accuracy improves. This is why better models have not dissolved the discomfort the surveys record.

4 The supervised rung and why it dominates the middle

Now add the supervised rung. Review cuts the error to $e(x)(1-\rho d)$. Because a clinician remains accountable, it also removes part of the aversion. Write the supervised aversion as

$$\pi_S(x,\rho) \;=\; \big(1-\beta\rho\big)\,\pi_A(x), \qquad \beta\in(0,1], \tag{3}$$

Full review ($\rho=1$) removes a fraction β of the aversion. Set the price $p_S\in(p_A,p_H)$ to reflect the clinician time that review consumes. The claim that a human in the loop both lowers error and lowers aversion is the one the behavioral literature supports most directly. Giving people the power to adjust an algorithm's output, even slightly, sharply raises their willingness to use it [Dietvorst et al., 2018]. And resistance to medical AI is muted when the machine supports rather than replaces the clinician [Longoni et al., 2019].

Proposition 2 (Supervision widens acceptance)**.** *Fix $\rho>0$, and suppose supervision is still accepted at the point where autonomy ceases to be, $\Delta V_S(\hat{x}_D)>0$. Then the supervised acceptance threshold $\hat{x}_S$, defined by $U_S(\hat{x}_S)=U_H(\hat{x}_S)$, satisfies $\hat{x}_S>\hat{x}_D$, and there is a middle interval of complexities on which the supervised rung yields strictly higher utility than both working alone and acting alone, so a patient offered all three chooses supervision there.*

Proof. The supervised-versus-human gap is

$$\Delta V_S(x)=\kappa\,\sigma(x)\big[e_H(x)-e(x)(1-\rho d)\big]+(p_H-p_S)-(1-\beta\rho)\pi_A(x),$$

which single-crosses by the same structure as ΔV and so has a unique threshold $\hat{x}_S$. We do not need $\Delta V_S>\Delta V$ everywhere, and in fact near $x=0$ the quality and aversion gaps are small and only the extra review price $-(p_S-p_A)$ survives, so $\Delta V_S<\Delta V$ there. What is needed is only $\Delta V_S(\hat{x}_D)>0$. Evaluating at $\hat{x}_D$, where $\Delta V=0$,

$$\Delta V_S(\hat{x}_D)=\kappa\,\sigma(\hat{x}_D)\,e(\hat{x}_D)\,\rho d \;+\; \beta\rho\,\pi_A(\hat{x}_D) \;-\; (p_S-p_A),$$

the error-detection gain plus the aversion relief minus the extra review price; when this is positive, single crossing of ΔV_S gives $\hat{x}_S>\hat{x}_D$. For the middle interval, $U_S-U_A=\kappa\sigma e\rho d+\beta\rho\pi_A-(p_S-p_A)$ is positive once x is large enough that the error-and-aversion gain outweighs the review price, while $U_S-U_H=\Delta V_S>0$ for $x<\hat{x}_S$; their intersection is the stated interval, nonempty exactly when $\Delta V_S(\hat{x}_D)>0$. □

Proposition 2 says the drafting-under-review rung is not a compromise. It is the efficient modality across a wide middle of medicine, because it buys down error and aversion together at the cost of some clinician time. This is the formal counterpart of what the market has chosen. The commercial reward has gone to ambient and drafting tools that sit under a clinician who signs, and physician adoption has followed. The regulator has drawn its central line in the same place. Software is treated as low-risk decision support rather than a regulated device only when the clinician can independently review the basis for its recommendation, and so does not rely on it primarily [U.S. Food and Drug Administration, 2022]. The autonomous rung was sold across all complexities as an AI doctor, and it is the one that failed in the market. That is the behavior Proposition 2 predicts for a product priced at p_A but accepted only on $x<\hat{x}_D$.

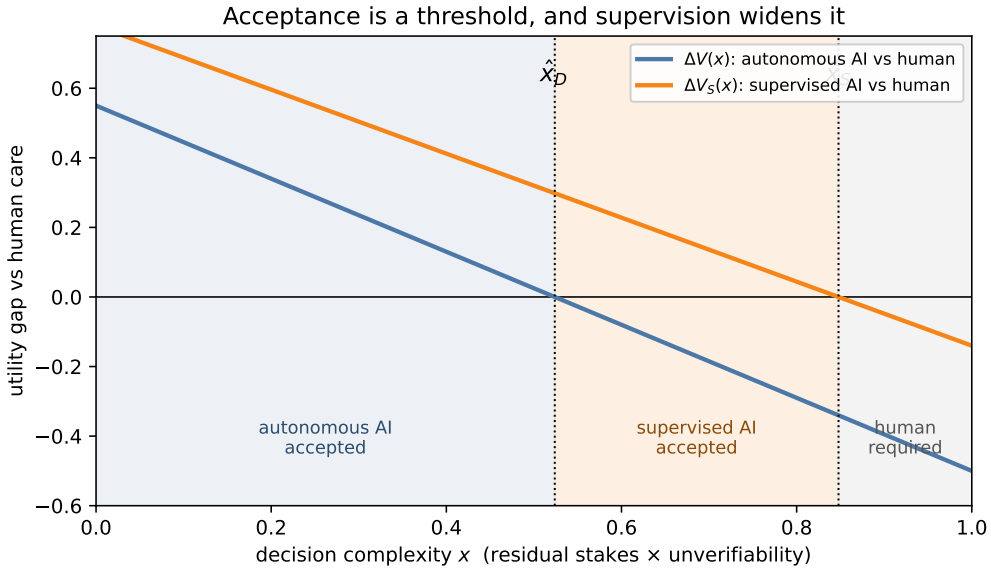


Figure 1: Patients accept a modality while its utility gap over human care is positive. The autonomous rung is accepted only up to $\hat{x}_D$. The supervised rung lowers both error and aversion, so it is accepted up to $\hat{x}_S > \hat{x}_D$ and wins the middle band of complexity. Above $\hat{x}_S$ patients want the human.

5 Supply: optimal oversight and the risk tiers

Turn to the clinic. It chooses how much to review. It trades the cost of clinician time against the harm that review prevents. Let $c > 0$ be the shadow price of a unit of clinician review time. It is high because that time is scarce [Porter et al., 2023, Association of American Medical Colleges, 2024]. Let $\tau(x)$ be the time a review of a task of complexity x takes. For review intensity $\rho \in [0, 1]$ the expected social cost of a task is

$$C(\rho; x) \;=\; \underbrace{c\,\tau(x)\,\rho}_{\text{review cost}} \;+\; \underbrace{\kappa\,\sigma(x)\,e(x)\,(1-\rho d)}_{\text{expected residual harm}}. \tag{4}$$

This is linear in ρ, so the optimum is at a corner. Which corner depends on whether the harm a marginal unit of review prevents exceeds its cost.

Proposition 3 (Oversight is a cutoff rule)**.** *The cost* (4) *is minimized by full review,* $\rho^\star(x) = 1$*, when*

$$\kappa\,\sigma(x)\,e(x)\,d \;\geq\; c\,\tau(x), \tag{5}$$

and by no review, $\rho^\star(x) = 0$*, otherwise. If* $\sigma(x)e(x)/\tau(x)$ *is increasing in* x*, there is a single safety threshold* $\hat{x}_O$ *such that every task above it is reviewed and every task below it is released. A tiered policy, review all tasks above an upper band, spot-check a middle band at an intensity that equates expected marginal harm to cost, and release a lowest band, is optimal when review capacity is constrained or when* $e(x)$ *is uncertain.*

Proof. Since $\partial C/\partial\rho = c\tau(x) - \kappa\sigma(x)e(x)d$ is constant in ρ, C is minimized at $\rho = 1$ when this derivative is negative, i.e. when (5) holds, and at $\rho = 0$ when it is positive; ties are immaterial. Under the monotonicity of $\sigma e/\tau$ the inequality (5) switches once as x rises, defining $\hat{x}_O$. With a

capacity constraint $\int \rho(x)\,dF(x) \le R$, the Lagrangian adds a shadow price to c, the same cutoff structure holds at the higher effective cost, and an interior ρ appears only at the marginal band, which is the spot-check tier; the uncertainty case is taken up in Proposition 7. □

Proposition 3 explains why real oversight looks the way it does. Consider a deployment that sorts patients into tiers: halt, specialist clearance, mandatory review, and a standard path eligible for spot-check release. It lets only the lowest-risk tier move off full review, and only after the AI has agreed with clinicians often enough. That deployment is implementing the cutoff (5). It reviews where $\sigma(x)e(x)$ is high and releases where it is low. The bands are a discretization of a monotone rule, not an arbitrary list.

6 The coincidence of the two frontiers

The demand threshold $\hat{x}_D$ and the safety threshold $\hat{x}_O$ are different objects. One is about what patients will accept, the other about what a clinic should allow. But they respond to the same primitive. In the maintained regime $\hat{x}_D > x^\star$ both thresholds fall as the residual stakes $\sigma(x)$ rise. The patient wants a human sooner where σ is high, because aversion bites hardest where the case is least verifiable. The clinic reviews sooner where σ is high, because that is where an uncaught error is most costly.

Proposition 4 (Coincidence and the wedge)**.** *In the regime $\hat{x}_D > x^\star$, holding the other primitives fixed, an increase in $\sigma(x)$ lowers $\hat{x}_D$ and $\hat{x}_O$ together, the safety threshold unconditionally and the demand threshold by Proposition 1. Hence the region where patients accept AI, $\{x < \hat{x}_D\}$, and the region where releasing it is safe, $\{x < \hat{x}_O\}$, move in the same direction and overlap on $\{x < \min(\hat{x}_D, \hat{x}_O)\}$. The two need not coincide exactly, and the ordering of the thresholds names two failures:*

(i) if $\hat{x}_D < \hat{x}_O$, there is an interval $(\hat{x}_D, \hat{x}_O)$ of excess aversion*, where patients refuse AI that is in fact safe to release;*

(ii) if $\hat{x}_D > \hat{x}_O$, there is an interval $(\hat{x}_O, \hat{x}_D)$ of excess trust*, where patients would accept AI that is not yet safe to release.*

Introducing the supervised rung with intensity ρ raises the acceptance threshold toward $\hat{x}_S$ and lowers the effective error to $e(1-\rho d)$, which widens the safe-release threshold as well, shrinking both intervals. In the limit $\beta\rho \to 1$ and $\rho d \to 1$ the two frontiers meet.

Proof. That $\hat{x}_D$ falls as σ rises holds in the regime $\hat{x}_D > x^\star$ by Proposition 1; that $\hat{x}_O$ falls as σ rises follows unconditionally from (5), whose left side is increasing in σ, so the crossing moves left. The two intervals in (i) and (ii) are the two possible orderings of the thresholds, and they are disjoint by construction. For the last claim, Proposition 2 gives $\hat{x}_S > \hat{x}_D$, and substituting $e(1-\rho d)$ for e in (5) raises the safety threshold because the left side falls; as $\beta\rho \to 1$ the supervised aversion vanishes and as $\rho d \to 1$ the supervised error vanishes, so the acceptance and safety conditions both reduce to statements about the human rung and the thresholds coincide. □

Proposition 4 sets up the paper's organizing question, but it is not yet the answer. Stated at this level the coincidence is close to mechanical. If both thresholds are monotone in the same index, of course they move together. Because the same residual-stakes index drives both sides, the market tends to settle where demand and safety agree. That is AI on the routine and reversible, and a human on the consequential and unverifiable. The surveys and the market then stop looking

contradictory. Majorities reject AI deciding their consequential care, and the market rewards AI drafting their routine care. These are the two ends of one frontier. The wedge is where the interesting policy lives. Excess aversion means patients refuse safe automation. It is the case for disclosure and for the assist-not-replace framing that is known to raise acceptance. It is a demand problem, not a safety problem. Excess trust means patients accept unsafe automation. It is the case for mandatory oversight and for the regulator's independent-review line. It is the Babylon case: a direct-to-consumer AI doctor accepted by some patients on tasks that were not safe to automate, whose triage safety was publicly questioned and which then failed commercially [TechCrunch, 2023]. The human in the loop is valuable precisely because it does not treat demand and safety as a trade-off. By raising acceptance and lowering error at once, it pulls both frontiers the same way. But the real content is in where the two frontiers separate. To locate that we have to stop bundling stakes with verifiability. That is the next section, and it is where the argument earns its keep.

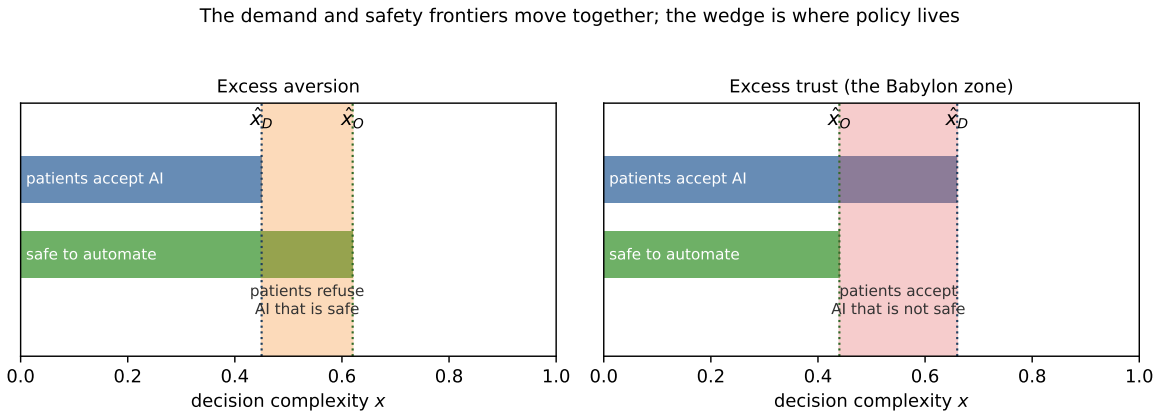


Figure 2: The demand threshold $\hat{x}_D$ and the safety threshold $\hat{x}_O$ are both driven by residual stakes, so they move together and largely overlap. Where they differ, the ordering names the failure. Excess aversion is patients refusing AI that is safe (left). Excess trust is patients accepting AI that is not safe, the Babylon zone (right). The supervised rung shrinks both wedges at once.

7 Accuracy is not verifiability: why better AI can demand more oversight

Up to here we folded two different things into the single index x. One is how much is at stake in a decision. The other is how easily anyone can tell whether it was right. For most of the argument that bundling is harmless, because the two rise together. But the sharpest results live exactly where they come apart, so we now separate them. A task is a pair (σ, v). The first is residual stakes σ, as before. The second is *verifiability* $v \in [0, 1]$, how readily the quality of the decision can be checked, by the patient or by a reviewing clinician. Medical care is a credence good precisely because v is often low [Arrow, 1963, Dulleck and Kerschbamer, 2006].

Verifiability changes what a reviewer can do. A human catches a share $d(v)$ of the AI's errors, with d increasing, $d(0) = 0$ and $d(1) = 1$. You can only catch what you can check. This splits the human's value into two channels that point in opposite directions in v, and that is the whole story. The *detection* channel catches AI mistakes and is worth $d(v)$, so it is strong exactly where

the task is verifiable. The *accountability* channel substitutes the human's own judgment and bears responsibility for it. It is what dissolves the patient's aversion [Longoni et al., 2019, Dietvorst et al., 2018], and it needs no verifiability at all.

Write the harm that reaches a patient as the *undetected harm*, and be careful about which rung carries a reviewer. Under autonomy no one checks the decision, so the harm is simply σe, and verifiability does not enter it. Verifiability enters only through the human. Under supervision the reviewer removes the share $d(v)$ they can catch, leaving $\sigma e(1 - d(v))$. Under a human deciding alone the harm is σe_H. So on a verifiable task supervision is a genuine safety technology. As $v \to 0$ the supervised harm $\sigma e(1 - d(v)) \to \sigma e$, and supervision stops reducing harm at all. A signature on a draft that no one can check is accountability, not safety. That is the economic content of the trial finding that a clinician given a strong model did no better than the model alone [Goh et al., 2024].

It is tempting to conclude that the safe-to-automate set is ordered by $\sigma(1 - d(v))$. That is the wrong accounting. Autonomy carries no reviewer, so its harm is σe, ordered by stakes alone. The channel through which verifiability really orders the rungs is learning. The planner does not know e. It estimates e from observed outcomes, and an outcome informs that estimate only when it can be verified. On a verifiable task the planner can confirm e is low enough to release. On an unverifiable one it never can, so a risk-averse planner keeps a human whatever the point estimate of e.

Proposition 5 (Verifiability governs the rung, through detection and learning)**.** *With autonomy harm σe, supervised harm $\sigma e(1 - d(v))$, and human-alone harm σe_H:*

(i) supervision lowers harm only on verifiable tasks, since its benefit $\sigma e\, d(v) \to 0$ as $v \to 0$; on an unverifiable task supervision is rubber-stamping, and the only safety lever is a human deciding, so the live choice there is autonomy at σe against a human at σe_H, not drafting under review;

(ii) because e is identified only from verifiable outcomes, autonomy can be earned*, shown safe enough to release, only where v is high; on a low-v task its safety cannot be established, so a risk-averse planner keeps the human whatever the point estimate of e;*

(iii) the order in which tasks can leave human control therefore runs by verifiability, not by stakes: a high-stakes but verifiable task can be released once it is learned safe, while a low-stakes but unverifiable task cannot, because its safety can never be shown.

Proof. (i) Autonomy harm σe has no reviewer and so does not depend on v; the supervised benefit is $\sigma e - \sigma e(1 - d(v)) = \sigma e\, d(v)$, which vanishes as $d(v) \to 0$, that is as $v \to 0$, leaving only autonomy against the human σe_H. (ii) Let the planner release only when its posterior that $\sigma e \leq \bar{\theta}$ exceeds a threshold fixed by its risk aversion; by the updating in Proposition 7, that posterior moves only on verified outcomes, so at $v = 0$ it never leaves the prior and the threshold is never met. (iii) Combining the two, a verifiable task reaches release once learning carries its posterior past the threshold, while an unverifiable task of any stakes does not, so the release order is governed by v, and a high-σ high-v task can precede a low-σ low-v one. □

This cuts against the severity-ordered autonomy ladders common in clinical-AI practice, which sort what to automate by how dangerous the condition is [International Medical Device Regulators Forum, 2014]. It does not cut against the regulator. The regulator's line is already a verifiability test. Non-device decision support is defined by whether the clinician can independently review the basis for the recommendation [U.S. Food and Drug Administration, 2022], which is exactly v. That

a model's accuracy can be established only where its outputs can be checked is also the premise of the scalable-oversight literature [Amodei et al., 2016, Bowman et al., 2022]. Our contribution is not that observation. It is carrying the observation into a model with a demand side and reading off where acceptance and safety part.

The counterintuitive dynamics follow once the AI improves, and they survive putting the patient's aversion back in. A patient accepts autonomy over a human when the price saving and the perceived quality gain outweigh the aversion:

$$(p_H - p_A) \;+\; \kappa\,\sigma\,[\,e_H - \hat{e}\,] \;\geq\; \pi_A(v), \tag{6}$$

Here $\hat{e}$ is the accuracy the patient perceives, and $\pi_A(v)$ is the aversion, which Assumption 3 makes larger on less verifiable tasks. Safety is the separate condition on the true autonomy harm, $\sigma e(v) \leq \bar{\theta}$. Accuracy is benchmarked where it can be measured, on verifiable tasks. So as the model improves, $\hat{e}$ falls everywhere while the true low-v error $e(v)$ lags.

Proposition 6 (Better AI can widen the excess-trust wedge)**.** *Suppose patients judge autonomy through the acceptance condition (6) on perceived accuracy $\hat{e}$, while true safety depends on $e(v)$. As the model improves, $\hat{e}$ and the high-v error fall together and the low-v error lags. Then:*

(i) acceptance expands on any task where the fall in perceived harm plus the price saving comes to exceed the aversion $\pi_A(v)$;

(ii) the safe region $\{\sigma e(v) \leq \bar{\theta}\}$ expands only where v is high, because $e(v)$ falls only there;

(iii) on low-v tasks where price or convenience is large enough to clear the aversion, acceptance expands while safety does not, so the excess-trust wedge grows there as the model improves.

The aversion of Assumption 3 works against the wedge, so it grows only where cheap, convenient autonomy overcomes the patient's reluctance. That is the direct-to-consumer segment, and it is where the market has already produced the failure.

Proof. The acceptance set is $\{(p_H - p_A) + \kappa\sigma(e_H - \hat{e}) \geq \pi_A(v)\}$, whose left side rises as $\hat{e}$ falls, so lowering $\hat{e}$ enlarges it wherever the inequality can be met, giving (i). The safe set $\{\sigma e(v) \leq \bar{\theta}\}$ enlarges only where $e(v)$ falls, that is at high v, giving (ii). The wedge is the accepted-but-unsafe set; on a low-v task where a large $p_H - p_A$ or $\kappa\sigma(e_H - \hat{e})$ overcomes $\pi_A(v)$, the acceptance inequality holds while $\sigma e(v) > \bar{\theta}$ persists because $e(v)$ has not fallen, and the measure of such tasks grows as $\hat{e}$ drops further below $e(v)$, giving (iii). □

This is the result the paper is built to deliver, and it cuts against the intuition that improving AI steadily reduces the need for people. It does reduce that need where the work is checkable. Where the work is not checkable, better average AI is precisely what lures unverifiable, high-stakes decisions into autonomous use. There the right response is not more review, which cannot catch what no one can check. It is to keep the human deciding, rung H rather than rung S. The direct-to-consumer "AI doctor" is the limiting case. It is marketed on benchmark accuracy, deployed on exactly the unverifiable complaints where that accuracy cannot be checked, and failing in the way Proposition 6 describes. The collapse of a symptom-checker-led telehealth unicorn into bankruptcy illustrates it [TechCrunch, 2023]. The nearest economic treatment is Ide and Talamàs [2025], who study autonomous versus supervised AI in a knowledge hierarchy where humans handle exceptions. The delegation-and-oversight tradeoff is the one Aghion and Tirole [1997] formalize. Neither carries a verifiability axis distinct from accuracy, and neither has a demand side. So the result that better accuracy can raise the optimal reliance on a human decision-maker is, to our knowledge, new.

8 Dynamics: earning autonomy

The safety rule (5) treats the AI's error rate $e(x)$ as known. It is not. A clinic learns it by watching the AI's decisions against its clinicians', the concordance that deployments record. This turns oversight into a sequential decision under a safety constraint. Reviewing a case both prevents harm today and reveals a signal about $e(x)$. Releasing a case saves clinician time but forecloses the learning that would justify releasing more. We make this precise for one complexity band and solve it.

8.1 A learning-and-oversight problem

Fix a band with residual stakes σ, per-case review cost $k \equiv c\,\tau$, detection rate $d(v)$, harm scale κ, and verifiability v. The true error rate $\theta \equiv e$ is unknown, with a Beta prior of mean μ_0. Reviewing a case produces a signal about θ, but how informative that signal is depends on v. On a fully verifiable task the reviewer observes whether the AI was right, a clean Bernoulli(θ) draw. On an unverifiable task the reviewer observes only whether the AI and the clinician agreed. That conflates the AI's error with the clinician's own error e_H, so it is a noisy proxy for θ. Write the signal precision as $\lambda(v)$, increasing in v, with $\lambda(1)$ full and $\lambda(0)$ carrying no information about θ beyond the prior. The posterior mean μ_t is a bounded martingale that concentrates at a rate increasing in $\lambda(v)$.

The clinic is risk-averse. It releases a band only when it is confident the band is safe, that is when the posterior probability $\mathbb{P}(\sigma\theta \le \bar{\theta})$ exceeds $1-\alpha$. For the Beta posterior that is a cutoff $\bar{\mu}$ on the mean. Each period it either reviews ($\rho = 1$) at cost k and expected harm $\kappa\sigma\mu_t(1-d(v))$, or releases ($\rho = 0$) at expected harm $\kappa\sigma\mu_t$. A released band still yields the sparse after-the-fact signal that, by Assumption 1, surfaces an error with the later-detection probability ϕ. Treating release as absorbing is an approximation for the leading-order cutoff. Because it ignores the option to return, it understates the value of releasing and so biases the computed $\bar{\mu}$ downward, toward too much reviewing. The leading-order Bellman equation is

$$V(\mu) = \min\Big\{ \underbrace{\tfrac{\kappa\sigma\mu}{1-\delta}}_{\text{release}} \;,\; \underbrace{\kappa\sigma\mu(1-d(v)) + k + \delta\,\mathbb{E}\big[V(\mu') \mid \mu\big]}_{\text{review}} \Big\}, \tag{7}$$

where μ' is the one-step posterior update under review.

Proposition 7 (Earning autonomy)**.** *With the chance-constrained release rule and the v-dependent signal precision $\lambda(v)$:*

(i) learning concentrates the posterior at a rate increasing in v, so autonomy is earned faster on more verifiable bands; at $v = 0$ the review signal does not identify θ and the posterior never tightens, so the band is never released, whatever its stakes;

(ii) among bands that can be identified ($v > 0$), the release cutoff $\bar{\mu}$ is decreasing in the residual stakes σ, so low-σ bands clear the bar first and on the least data, and a band whose σ is large enough that $\bar{\mu}$ falls below any attainable posterior mean is never released; the order of release runs by verifiability first, whether safety can be learned at all, and by stakes second, how low the bar sits;

(iii) under release the ϕ channel makes silence informative, since no surfaced error is evidence of low θ and walks μ further down, so a single detected error triggers rollback early, when μ sits just below $\bar{\mu}$, but after a long quiet spell a single error no longer clears $\bar{\mu}$; the optimal rollback is therefore not one strike but k strikes, with k growing in the time since release.

Proof. (i) A Beta posterior updated by a signal of precision $\lambda(v)$ has posterior variance falling at a rate increasing in $\lambda(v)$; at $\lambda(0) = 0$ the posterior equals the prior for all t, so $\mathbb{P}(\sigma\theta \leq \bar{\theta})$ never rises to $1-\alpha$ and release never occurs. (ii) For $v > 0$ the chance-constrained cutoff solves $\mathbb{P}(\sigma\theta \leq \bar{\theta} \mid \mu) = 1-\alpha$; since the harm is $\sigma\theta$, a larger σ tightens that event and lowers the admissible $\bar{\mu}$, and if the required $\bar{\mu}$ lies below every attainable posterior mean the band is never released. With (i), identifiability (a function of v) gates whether release is ever reached and σ sets how far learning must go, the stated lexicographic order. (iii) Under the ϕ channel a released band that produces no surfaced error has a posterior that drifts down, since absence of detected error is evidence of low θ; μ falls below $\bar{\mu}$ by a growing margin, and the upward jump from a single detected error clears $\bar{\mu}$ only while that margin is small, that is soon after release, so the number of detected errors needed to trigger rollback rises with time since release. □

Proposition 7 matches the structure of careful real deployments without being told to. Consider a phased rollout. It begins with full review of every decision. It lets only the lowest-risk tier move to a sampled audit, and only after the AI has agreed with clinicians at a high rate for a sustained period. It keeps the least verifiable and highest-risk tiers under full review permanently. And it returns a tier to full review after a run of adverse signals, tolerating fewer the more recently autonomy was granted. That rollout is implementing the three parts of the proposition. The reason to review a decision is not only to catch its error today. Early on it is also to learn whether the machine can be trusted with tasks like it tomorrow. That learning is possible at all only where the work is verifiable, and it is cheapest where the residual stakes are also low.

9 What the theory leaves for measurement

The model is built to be tested, and it says what to measure. It makes two kinds of prediction, and the second is the one the literature has no evidence on. The first is the baseline ordering. Residual stakes crossed with verifiability sort both patient acceptance and safe automation, so acceptance should fall and optimal oversight should rise along that single index. The literature has measured acceptance task by task, finding it high for diagnosis and low for triage. But it has not mapped a continuous stakes axis, and it has almost no evidence on how acceptance moves as patients accumulate experience or witness a single error. A discrete-choice experiment would fix this. Vary the rung, the residual stakes, the reversibility, the price, and the presence of a human reviewer. That would estimate $\hat{x}_D$ and the willingness to pay for a human directly, and would locate the excess-aversion and excess-trust wedges where disclosure and oversight have work to do.

The second prediction is the separation of accuracy from verifiability. Testing it needs a manipulation no prior study has run. Existing work moves perceived uniqueness or stakes. None moves *verifiability* while holding stakes fixed. Tell one arm that the decision's outcome will be objectively checkable afterward, by anyone, and tell another that it will never be verifiable, with the stakes held constant. That should shift the acceptance threshold on its own, which is the demand-side signature of Proposition 5. Note that this manipulates whether the work can be checked at all, not whether a particular clinician reviews it, which would instead move supervision. On the supply side the same index predicts that observed oversight intensity tracks verifiability over severity. That is testable. Code the roughly fifteen hundred FDA-cleared AI devices by the autonomy they were granted against the severity of the condition they address, and read clinician-override logs as a verifiability proxy. The sharpest and most falsifiable claim comes from Proposition 6. As models improve, the gap between accepted and safe automation should *widen* on low-verifiability tasks rather than close. A repeated cross-section of acceptance and adverse events as model quality rises would detect it. Pairing all of this with concordance data from a supervised deployment would

let the dynamics of Proposition 7 be estimated rather than assumed. We develop the design in a companion paper. The claim here is simple. The question worth measuring is not whether patients want a human or a machine. It is where on the ladder each decision belongs. The answer is written in the stakes and in whether anyone can check the work.

10 Conclusion

People are not of two minds about artificial intelligence in medicine. They are of one mind about two different things. They want a machine to take the routine, reversible work that a clinician would have done in a hurry. And they want a human to stay accountable for the consequential, unverifiable decisions whose quality they cannot check. The clinic trades scarce clinician time against the harm of an uncaught error, and it wants to draw its line in the same place. Both lines are drawn by the residual stakes of the decision, not by the sophistication of the model. So they move together. The equilibrium the market has already found, AI that drafts under a clinician who approves, is the one the theory predicts. The autonomous AI doctor failed not because the technology was poor. It failed because it was sold across the whole ladder when it belonged only on the bottom rung. The result that should change how people plan for better models is the one that runs against intuition. A machine's accuracy is proved on the decisions that can be checked, and then trusted on the ones that cannot. So each advance in AI quality can widen rather than close the gap between the care patients will accept and the care that is safe to automate, wherever cheap and convenient autonomy overcomes a patient's reluctance. On the hardest and least verifiable decisions the right response to a better model is therefore to keep a human deciding, not merely reviewing, since review cannot catch an error no one can check. The useful questions from here are quantitative. Where exactly do the thresholds fall for a given kind of care? How wide is the wedge between acceptance and safety? How quickly can a clinic earn the right to step back? Those are measurable, and they are the ones worth asking.

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